



# 5896 - 47Tuc: A Second Epoch to Extend the Mass-Rotation Relation to the Brown Dwarf Regime

Cycle: 3, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                               | <i>Institution</i>                        |
|-------------------------------------------|-------------------------------------------|
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| Dr. Massimo Griggio (CoI)                 | Space Telescope Science Institute         |
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## OBSERVATIONS

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i>               | <i>Observing Template</i> | <i>Science Target</i> |
|--------------------|--------------------|----------------------------|---------------------------|-----------------------|
| Observation Folder |                    |                            |                           |                       |
|                    | 1                  | NGC104-v123-GO2559         | NIRCam Imaging            | (1) NGC-104           |
|                    | 2                  | Residual Persistence Check | NIRCam Dark               | NONE                  |

## ABSTRACT

A recent investigation of the internal kinematics of a few globular clusters has provided the first observational evidence of a correlation between the rotational velocity of the stars around the cluster's center and the stars' masses. That more massive stars rotate more rapidly around the clusters' center than low-mass stars, was predicted theoretically.

The first observational detection marks a significant advancement in our exploration of the dynamic and kinematic evolution of globular clusters, but it is thus far based on a narrow stellar mass range and limited to masses larger than 0.65 $M_{\odot}$ . The full study of this trend and its exploitation to gather

new fundamental insights into the manifestations of the relaxation-driven dynamics in star clusters requires exploring this relation over a broader mass range.

Here we propose JWST observations to extend the investigation of this relationship into the lower mass range, reaching the mass domain of brown dwarfs.

We plan to observe an outer region of the globular cluster 47Tuc, previously studied by JWST, which revealed a dozen strong brown dwarf candidates.

The new JWST observations will serve two additional main objectives. First, we will increase the number of brown dwarf candidates within 47 Tuc, and second, we will validate their cluster membership by precisely measuring their proper motions.

Furthermore, we will explore the multiple population phenomenon on the lower main sequence of 47Tuc and investigate the behaviour of white dwarf cooling sequences in the infrared spectrum.

## **OBSERVING DESCRIPTION**

In this program, we are requesting JWST NIRCам and parallel NIRISS observations of an external region of 47 Tuc, which was already observed during Cycle 1 in GO-2560 (PI: Marino), in order to obtain precise photometric and astrometric information on low-mass objects ( $<0.1$  solar masses) that are members of the cluster.

The observations collected in Cycle 1 in July 2022 experienced technical problems, specifically a guide star error, resulting in the discarded data being made public. As a result, the observations will be rescheduled for June 2023, and a different field will be observed in accordance with the APT. Despite the problems affecting the images collected in July 2022, it is still possible to extract high-quality photometry and astrometry of the stars in the field. The colour-magnitude diagram obtained from this dataset reaches a magnitude limit of  $F322W2 \sim 26$ , including the region where cluster brown dwarfs are expected.

A dozen candidate brown dwarfs have been identified through the combination of JWST and HST data, with an overlap of less than 50%. The requested observations will help us to confirm their membership to 47 Tuc and discover new brown dwarfs. Additionally, these observations will provide us with precise proper motion measurements, enabling us to extend the study of the rotation-mass relationship in 47 Tuc to lower masses, encompassing the brown dwarfs' regime.

We plan to map the same region that was observed during Cycle 1. The inclusion of NIRISS fields will allow us to find new objects with masses below 0.1 solar masses and study the phenomenon of multiple populations in the lower main sequence of 47 Tuc. The observations will be dithered using two large dithers to cover gaps between detectors. These dithers will also enable us to extract PSF models, calibrate the geometric distortion of the camera, and measure the photometric zero points between detectors.

The first epoch observations have been matched with deep WFC3/IR catalogues obtained from GO-11677 (PI: Richer) data. However, even though these datasets reach the faint magnitude regime, the proper motions are not accurate enough to confirm the membership of the discovered candidate brown dwarfs or to perform a precise kinematic analysis. Therefore, a second epoch of observations with JWST will be required to distinguish between cluster members and field/SMC stars while providing high-quality proper motion measurements. The requested observations will be planned in such a way that the NIRISS fields will cover an already observed field by HST or JWST.

For NIRCам observations, we are requesting images in the ultra-wide filters F150W2 and F322W2, as they are the best options for deep observations of low-mass objects that span a wide range of temperatures. For NIRISS, we are requesting observations in F140M, which is well-suited for the detection of more massive brown dwarfs, primarily because of its sensitivity to the H<sub>2</sub>O band.

# Proposal 5896 - Targets - 47Tuc: A Second Epoch to Extend the Mass-Rotation Relation to the Brown Dwarf Regime

| Fixed Targets | #                                                                                                                                                                     | Name    | Target Coordinates             | Targ. Coord. Corrections                     | Miscellaneous |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------|----------------------------------------------|---------------|
|               | (1)                                                                                                                                                                   | NGC-104 | RA: 00 22 0.0000 (5.5000000d)  | Proper Motion RA: 5.25 mas/yr                |               |
|               |                                                                                                                                                                       |         | Dec: -72 05 20.00 (-72.08889d) | Proper Motion Dec: -2.529999915168446 mas/yr |               |
|               |                                                                                                                                                                       |         | Equinox: J2000                 | Epoch of Position: 2000                      |               |
|               | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Stellar Cluster<br>Description=[Globular star clusters] |         |                                |                                              |               |

# Proposal 5896 - Observation 1 - 47Tuc: A Second Epoch to Extend the Mass-Rotation Relation to the Brown Dwarf Regime

|                   |                                                                                                                                                                       |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------|-----------------|------------------|----------------------------------------------|-----------------|--------------------|--------------------|----------------------------------------|------------------------------|-----------------------------|
| Observation       | Proposal 5896, Observation 1: NGC104-v123-GO2559                                                                                                                      |                     |                                |                 |                  |                                              |                 |                    |                    |                                        | Wed Aug 27 17:00:08 GMT 2025 |                             |
|                   | Diagnostic Status: Warning                                                                                                                                            |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|                   | Observing Template: NIRCcam Imaging                                                                                                                                   |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|                   | Coordinated Parallel Template(s): NIRISS Imaging                                                                                                                      |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
| Diagnostics       | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                           |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|                   | (Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                           |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
| Fixed Targets     | #                                                                                                                                                                     | Name                | Target Coordinates             |                 |                  | Targ. Coord. Corrections                     |                 |                    | Miscellaneous      |                                        |                              |                             |
|                   | (1)                                                                                                                                                                   | NGC-104             | RA: 00 22 0.0000 (5.5000000d)  |                 |                  | Proper Motion RA: 5.25 mas/yr                |                 |                    |                    |                                        |                              |                             |
|                   |                                                                                                                                                                       |                     | Dec: -72 05 20.00 (-72.08889d) |                 |                  | Proper Motion Dec: -2.529999915168446 mas/yr |                 |                    |                    |                                        |                              |                             |
|                   |                                                                                                                                                                       |                     | Equinox: J2000                 |                 |                  | Epoch of Position: 2000                      |                 |                    |                    |                                        |                              |                             |
|                   | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Stellar Cluster<br>Description=[Globular star clusters] |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
| Template          | NIRCcam Imaging                                                                                                                                                       |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|                   | NIRISS Imaging                                                                                                                                                        |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|                   | Module: ALL                                                                                                                                                           |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|                   | Subarray: FULL                                                                                                                                                        |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
|                   | Target Placement: Module Gap                                                                                                                                          |                     |                                |                 |                  |                                              |                 |                    |                    |                                        |                              |                             |
| Mosaic            | Rows                                                                                                                                                                  | Columns             | Row Overlap %                  |                 | Column Overlap % |                                              | Row shift (deg) |                    | Column shift (deg) |                                        | Tile Order                   |                             |
|                   | 2                                                                                                                                                                     | 1                   | 10.0                           |                 | 10.0             |                                              | 0.0             |                    | 0.0                |                                        | DEFAULT                      |                             |
| Dithers           | #                                                                                                                                                                     | Primary Dither Type |                                | Primary Dithers |                  | Dither Size                                  |                 | Subpixel Positions |                    | Coordinated Parallel Subpixel Selector |                              | Dither Direct Images Primes |
|                   | 1                                                                                                                                                                     | FULLBOX             |                                | 6TIGHT          |                  |                                              |                 | 1                  |                    | NIRCcam Only                           |                              | NO_DITHERING                |
| Spectral Elements | NIRCcam Imaging                                                                                                                                                       | Short Filter        | Long Filter                    | Readout Pattern | Groups/Int       | Integrations/Exp                             |                 | Total Integrations | Total Dithers      | Total Exposure Time                    | Optional ETC ID              |                             |
|                   | 1                                                                                                                                                                     | F090W               | F444W                          | MEDIUM8         | 6                | 1                                            |                 | 6                  | 6                  | 3736.396                               |                              |                             |
| Spectral Elements | NIRISS Imaging                                                                                                                                                        | Filter              | Grism                          | Readout Pattern | Groups/Int       | Integrations/Exp                             |                 | Total Dithers      | Total Integrations | Total Exposure Time                    | Optional ETC ID              |                             |
|                   | 1                                                                                                                                                                     | F150W               |                                | NIS             | 14               | 1                                            |                 | 6                  | 6                  | 3671.975                               |                              |                             |

Proposal 5896 - Observation 1 - 47Tuc: A Second Epoch to Extend the Mass-Rotation Relation to the Brown Dwarf Regime

|                      |                                                                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | Sequence Visits , Non-interruptible<br>Aperture PA Range 341.92542306 to 341.92542306 Degrees (V3 342.0 to 342.0)<br>Visits Same PA<br>No Parallel Attachments<br>Persistence Risk<br><br>2 After 1 by 9.6 Hours to 2 Days, Exclusive Use Of Instrument |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Proposal 5896 - Observation 2 - 47Tuc: A Second Epoch to Extend the Mass-Rotation Relation to the Brown Dwarf Regime

|                      |                                                                                                                                                                                 |                 |                  |                  |                |                    |                        |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|------------------|----------------|--------------------|------------------------|------------------------------|
| Observation          | Proposal 5896, Observation 2: Residual Persistence Check<br>Diagnostic Status: Warning<br>Observing Template: NIRCam Dark<br>Comments: adding comment to trigger change checker |                 |                  |                  |                |                    |                        | Wed Aug 27 17:00:08 GMT 2025 |
| Diagnostics          | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                     |                 |                  |                  |                |                    |                        |                              |
| Template             | Module                                                                                                                                                                          | Subarray        | Science Template |                  | Occulting Mask |                    | No. of Output Channels |                              |
|                      | ALL                                                                                                                                                                             | FULL            | Imaging          |                  |                |                    | 4                      |                              |
| Spectral Elements    | #                                                                                                                                                                               | Readout Pattern | Groups/Int       | Integrations/Exp | Exposures/Dith | Total Integrations | Total Exposure Time    | Optional ETC ID              |
|                      | 1                                                                                                                                                                               | MEDIUM8         | 10               | 1                | 1              | 1                  | 1052.203               |                              |
| Special Requirements | 2 After 1 by 9.6 Hours to 2 Days, Exclusive Use Of Instrument                                                                                                                   |                 |                  |                  |                |                    |                        |                              |